Conceptual Design for EHR Systems

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What is Conceptual Design ?

Concepts are ideas needed to construct the software

- *Paragraphs* or *styles* for word processing

Concepts encapsulate the functionality needed, minimizing communication and increasing software reliability.

And concepts can be reused.

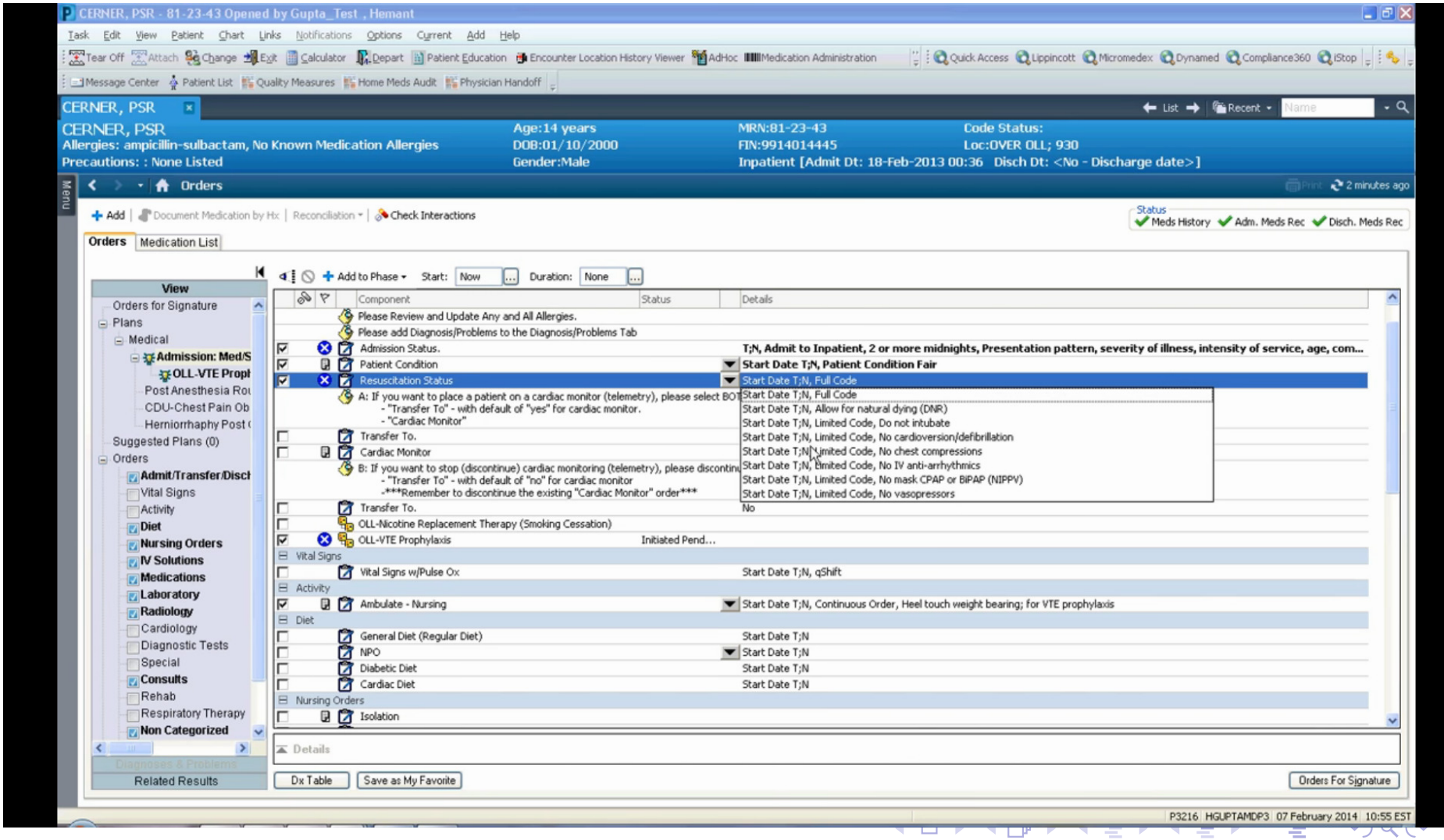
What are EHRs ?

Electronic Health Record systems manage patient and treatment data for hospitals and clinics.

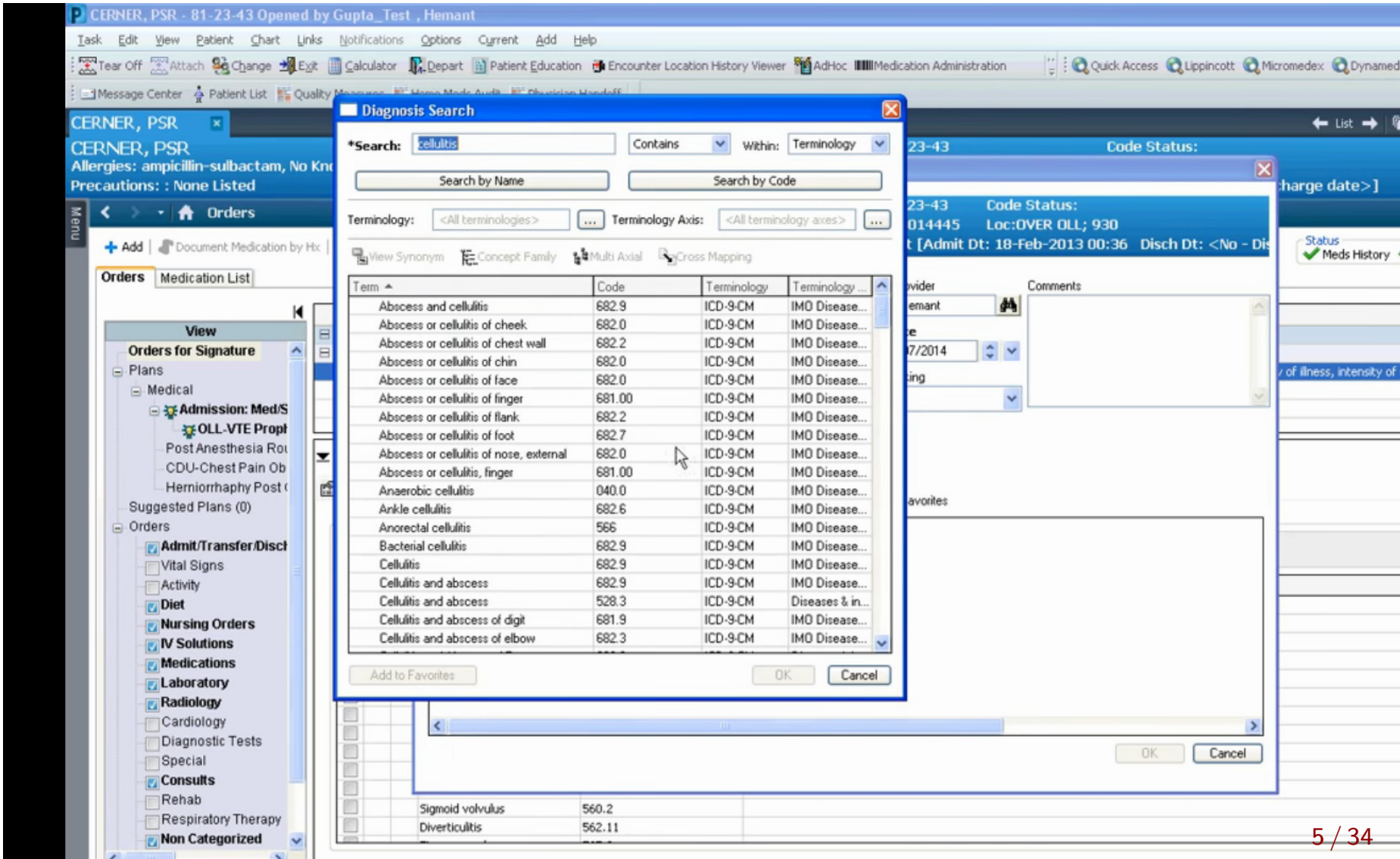
- Patient data
- Consultations
- Exams
- ...

Incredibly large and complex.

EHR Example


CERNER, PSR - 81-23-43 Opened by Gupta_Test , Hemant
CERNER, PSR
Allergies: ampicillin-sulbactam, No Known Medication Allergies
Precautions: : None Listed
Age:14 years
DOB:01/10/2000
Gender:Male
MRN:81-23-43
FIN:9914014445
Inpatient [Admit Dt: 18-Feb-2013 00:36 Disch Dt: <No - Discharge date>]
Orders
Orders
Medication List
Add to Phase
Start: Now
Duration: None
Component
Status
Details
Please Review and Update Any and All Allergies.
Please add Diagnosis/Problems to the Diagnosis/Problems Tab
Admission Status.
T;N, Admit to Inpatient, 2 or more midnights, Presentation pattern, severity of illness, intensity of service, age, com...
Patient Condition
Start Date T;N, Patient Condition Fair
Resuscitation Status
Start Date T;N, Full Code
Start Date T;N, Allow for natural dying (DNR)
Start Date T;N, Limited Code, Do not intubate
Start Date T;N, Limited Code, No cardioversion/defibrillation
Start Date T;N, Limited Code, No chest compressions
Start Date T;N, Limited Code, No IV anti-arrhythmics
Start Date T;N, Limited Code, No mask CPAP or BiPAP (NIPPV)
Start Date T;N, Limited Code, No vasopressors
Transfer To.
No
Cardiac Monitor
OLL-Nicotine Replacement Therapy (Smoking Cessation)
OLL-VTE Prophylaxis
Initiated Pend...
Vital Signs
Vital Signs w/Pulse Ox
Start Date T;N, qShift
Activity
Ambulate - Nursing
Start Date T;N, Continuous Order, Heel touch weight bearing; for VTE prophylaxis
Diet
General Diet (Regular Diet)
Start Date T;N
NPO
Diabetic Diet
Cardiac Diet
Nursing Orders
Isolation
Dx Table
Save as My Favorite
Orders For Signature
P3216 HGUPTAMDP3 07 February 2014 10:55 EST

EHR Example


CERNER, PSR - 81-23-43 Opened by Gupta_Test , Hemant
Diagnosis Search
*Search: cellulitis
Contains
Within: Terminology
Search by Name
Search by Code
Terminology:
Terminology Axis:
Term
Code
Terminology
Abscess and cellulitis 682.9 ICD-9-CM
Abscess or cellulitis of cheek 682.0 ICD-9-CM
Abscess or cellulitis of chest wall 682.2 ICD-9-CM
Abscess or cellulitis of chin 682.0 ICD-9-CM
Abscess or cellulitis of face 682.0 ICD-9-CM
Abscess or cellulitis of finger 681.00 ICD-9-CM
Abscess or cellulitis of flank 682.2 ICD-9-CM
Abscess or cellulitis of foot 682.7 ICD-9-CM
Abscess or cellulitis of nose, external 682.0 ICD-9-CM
Abscess or cellulitis, finger 681.00 ICD-9-CM
Anaerobic cellulitis 040.0 ICD-9-CM
Ankle cellulitis 682.6 ICD-9-CM
Anorectal cellulitis 566 ICD-9-CM
Bacterial cellulitis 682.9 ICD-9-CM
Cellulitis 682.9 ICD-9-CM
Cellulitis and abscess 682.9 ICD-9-CM
Cellulitis and abscess 528.3 ICD-9-CM
Cellulitis and abscess of digit 681.9 ICD-9-CM
Cellulitis and abscess of elbow 682.3 ICD-9-CM
Add to Favorites
OK
Cancel
Orders
Medication List
View
Orders for Signature
Plans
Medical
Suggested Plans (0)
Orders
Admit/Transfer/Disch
Vital Signs
Activity
Diet
Nursing Orders
IV Solutions
Medications
Laboratory
Radiology
Cardiology
Diagnostic Tests
Special
Consults
Rehab
Respiratory Therapy
Non Categorized
Sigmoid volvulus 560.2
Diverticulitis 562.11

What about EHRs ?

- EHRs could use a boost
- Conceptual design has been used so far for simpler types of software.
- Is it applicable for more complex types of software ? such as...
  - EHRs
  - ERPs
  - Manufacturing
  - Scientific
  - ...

Let's find out.

Types of Concepts

Currently Dejavu supports concepts such as:

- authentication
- comment
- event
- schedule
- ...

But EHRs have concepts such as:

- Consultation
- Doctor
- Patient
- Interview
- Diagnostic
- ...

Might have to write a few more concepts...

How do Doctors Work ?

1. Schedule
2. Consultations
   - Interviews
   - Previous consultations
   - Exams
3. Diagnostics
4. Conducts
5. Exams

All Doctors ???

1. EMS: Emergencies have no Schedule
2. Surgery: It has an additional *Procedure* step.
3. ...

1. Schedule
2. Consultations
   - Interviews
   - Previous consultations
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3. Diagnostics
4. Conducts
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The Interview Concept (1)

Name: interview

Purpose: to interview a subject on a set of prepared questions

Structure:

type: interview $\rightarrow$! interview-type

questions: interview-type $\rightarrow$ question

interviews: interview $\rightarrow$ question $\rightarrow$ answer

The Interview Concept (2)

Behaviors:

add-questions (t: interview-type, q: question)

t → q not in questions ⇒ questions += t → q

remove-questions (t: interview-type, q: question)

t → q in questions ⇒ questions -= t → q

list-questions (t: interview-type)

return questions.t

create (i: interview, t: interview-type, A: Answer, Q: Question)

for all (q in questions.t) ⇒ Q!q; A?a; interviews += i → q →

retrieve (i: interview-id)

return interviews.i

Tactic: if add-questions(t, q) and create (i, t) then

retrieve (i) will contain i → q → a

The Patient Concept

Name: patient
Purpose: to provide information about the patient and
answer doctors questions
Structure:
patients: set Patient
Behaviors:
create (p: Patient)
no patients.p ⇒ patients += d
remove (p: Patient)
patients.p ⇒ patients -= d
answer (p: Patient, Q: Question, A: answer)
for all Q?q ⇒ A!patient-answer(p, q)
Tactic: if create(p) then
can consultation.create (c, p, d, s) and
can answer (p, q, a)

A Consultation

```
read (d: Doctor, p: Patient, s: Slot);
{
      interview.create (c, doctor-specialty(d), Q, A) ||
      patient.answer (p, Q, A) ||
      {
            evidences = interview.retrieve(c);
            requests.show-results (p, d, lab-results);
            evidences += lab-results;
            consultation.show (p, previous-history);
            diagnostic.diagnose(c, evidences, previous-history, new-request
      } ||
      treatment.prescribe (c) ||
      request.add (c, new-requests, s);
}
```

Does it Work For All Doctors???

A bit generic.....

- Not all consultations serve the same purpose...
- Do we really understand the user ???
- From the user (doctors) perspective consultation is highly overloaded

Perhaps we should have a different consultation concept for each doctor ???

A Different View

A different consultation concept for each doctor ???

YES! If we change perspective

- Don't think of the software, think of the doctor
- What does he/she do ?
- Instead of organizing software functions, we should organize doctors activities...

Conceptual design for doctors, not software

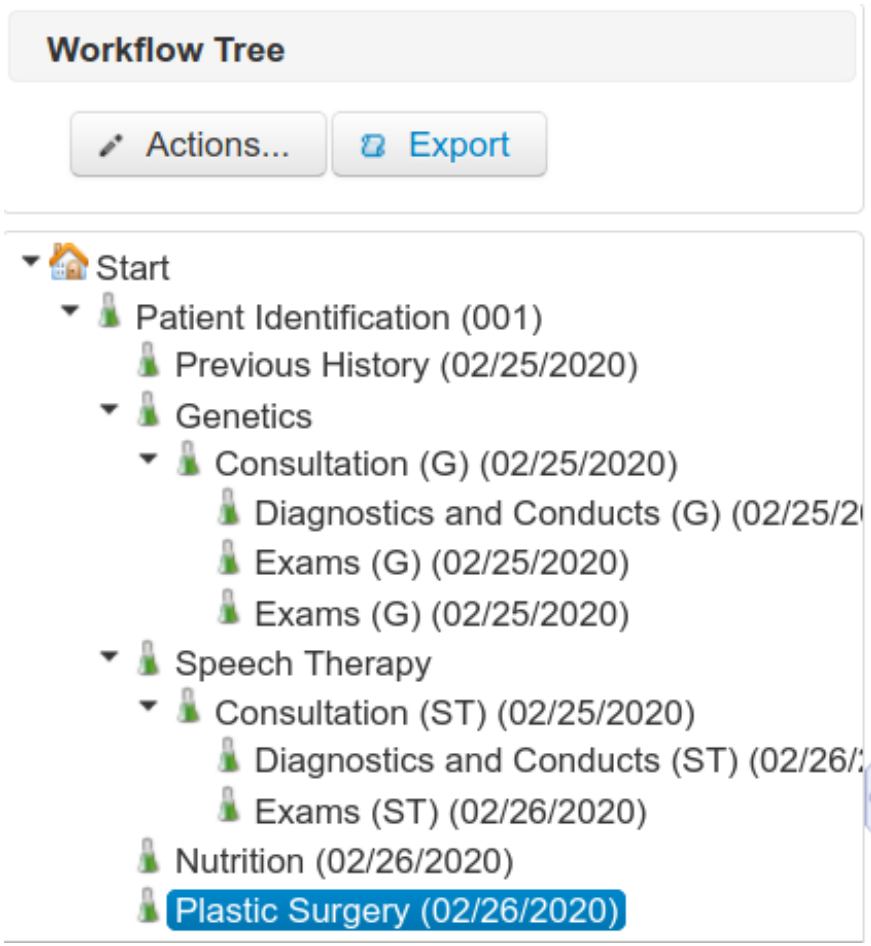
Role Based Software Development*

What if EHRs are *role based*?

- A different system for each role,
- Each user has a simple system,
- And the sum of all systems is the EHR.

*: Does this even exists ???

Complete System

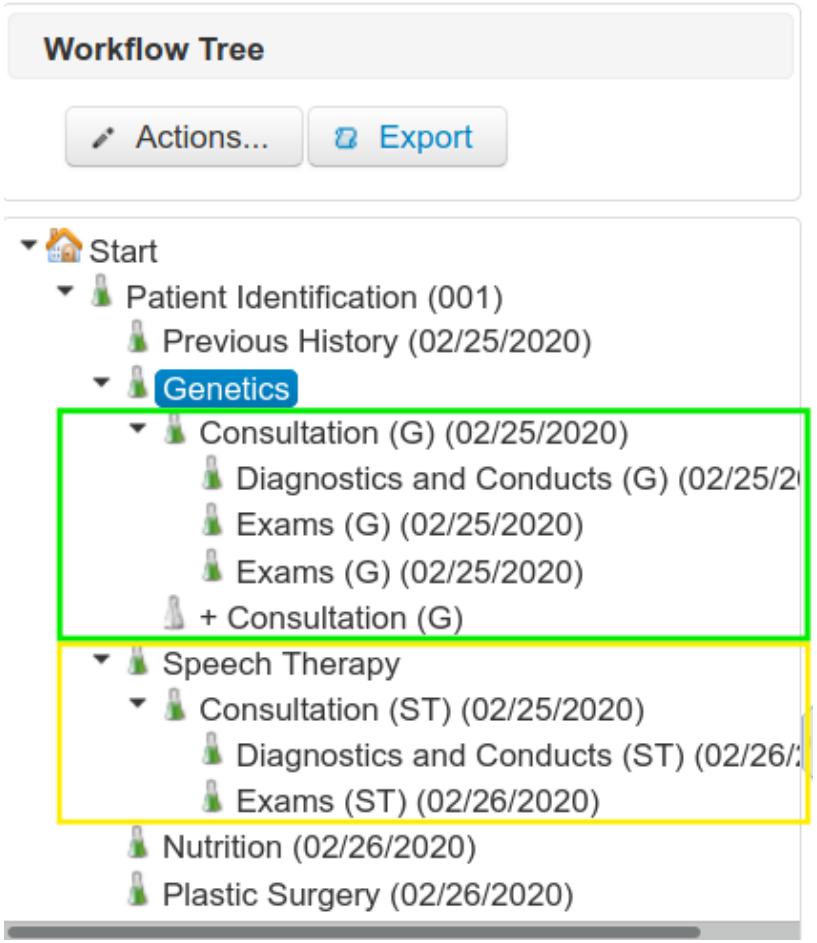


Roles

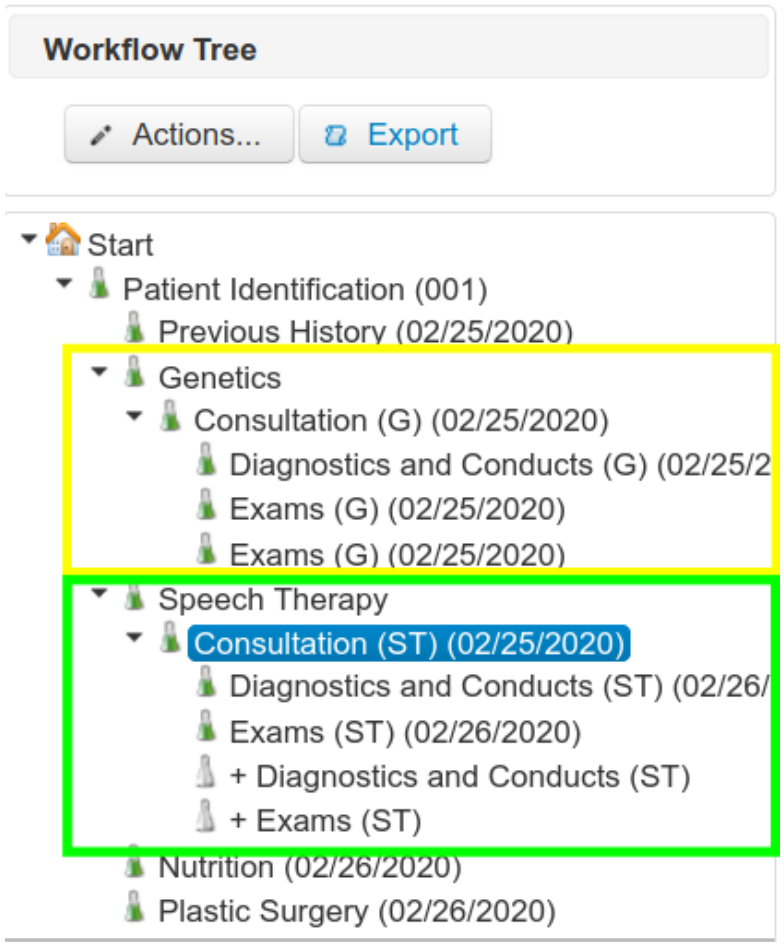
The most important roles are:

- Doctor:
  - Geneticist
  - Speech Therapist
  - Nutritionist
  - Plastic Surgeon
- Administrative Assistant
- Laboratory Technician

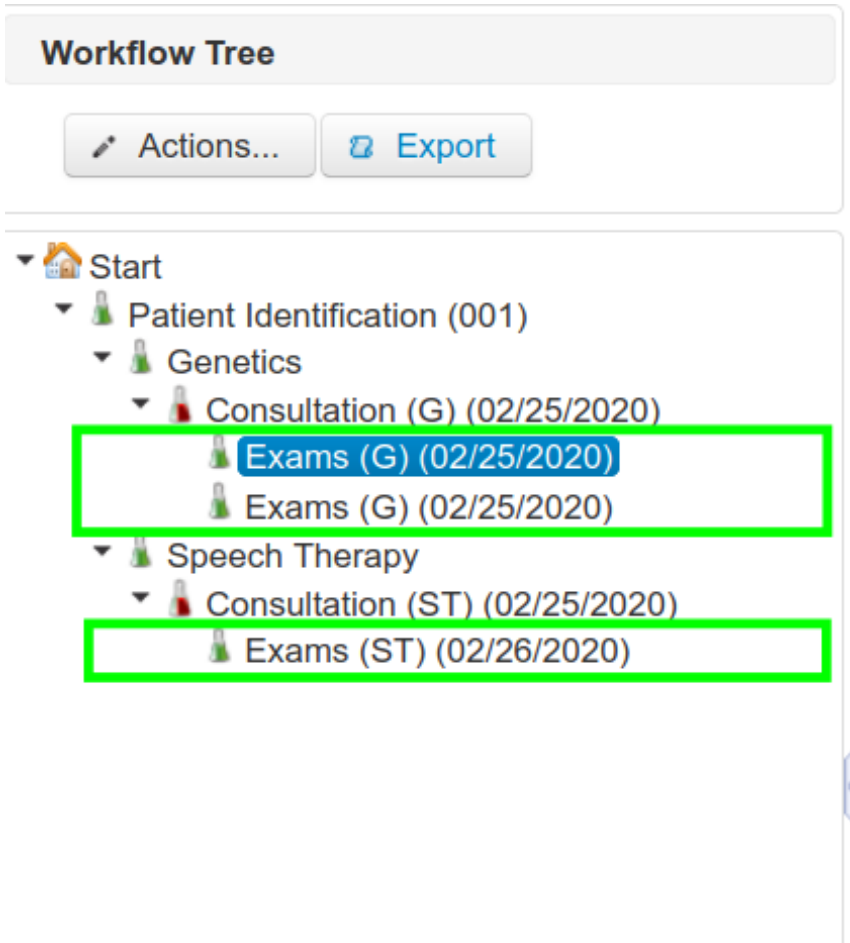
Geneticists View



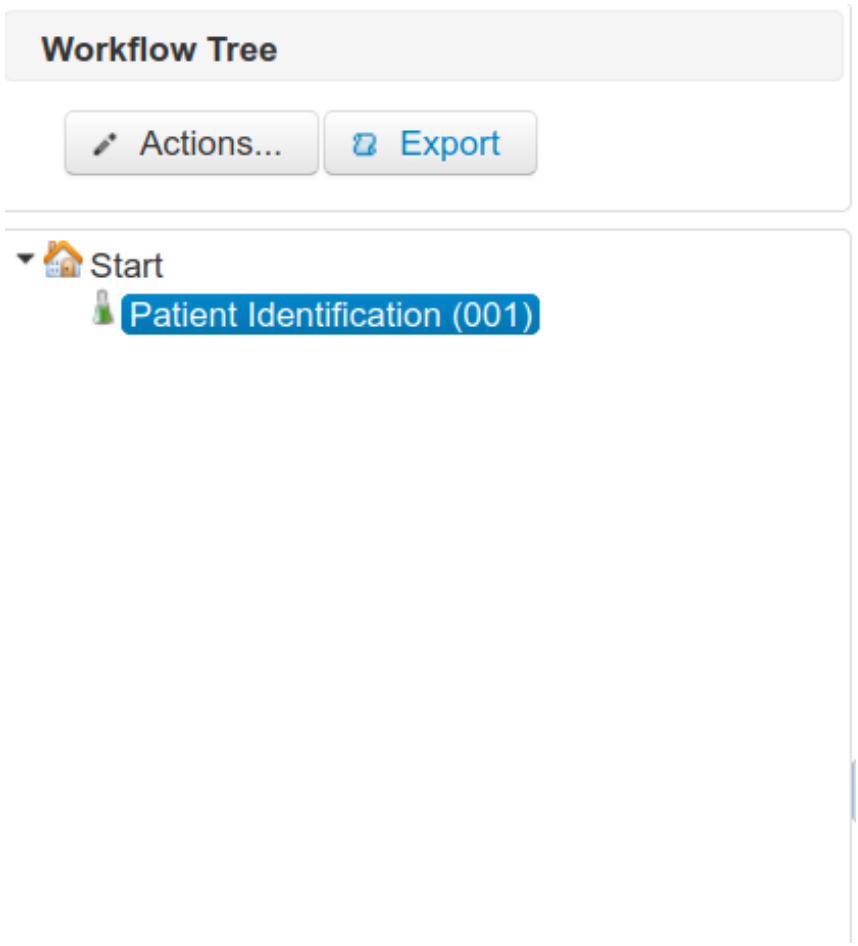
Speech Therapist View



Lab Technician View



Administrative Assistant View



Specialized Concepts

Isn't that a problem ? Do we violate design principles ?

- Instrumentality: one concept per purpose
- Familiarity: same concept for given purpose
- Integrity: composition preserves concepts

Not much reuse...

- No, but lots of fine grained compositionality.
- Avoiding conflicts between concepts and purposes

Conflicts between concepts and purposes

- A consultation may include an exam request
- Exam requests could be ECG/allergy test/...
- A cardiologist may request an ECG, but no allergy test

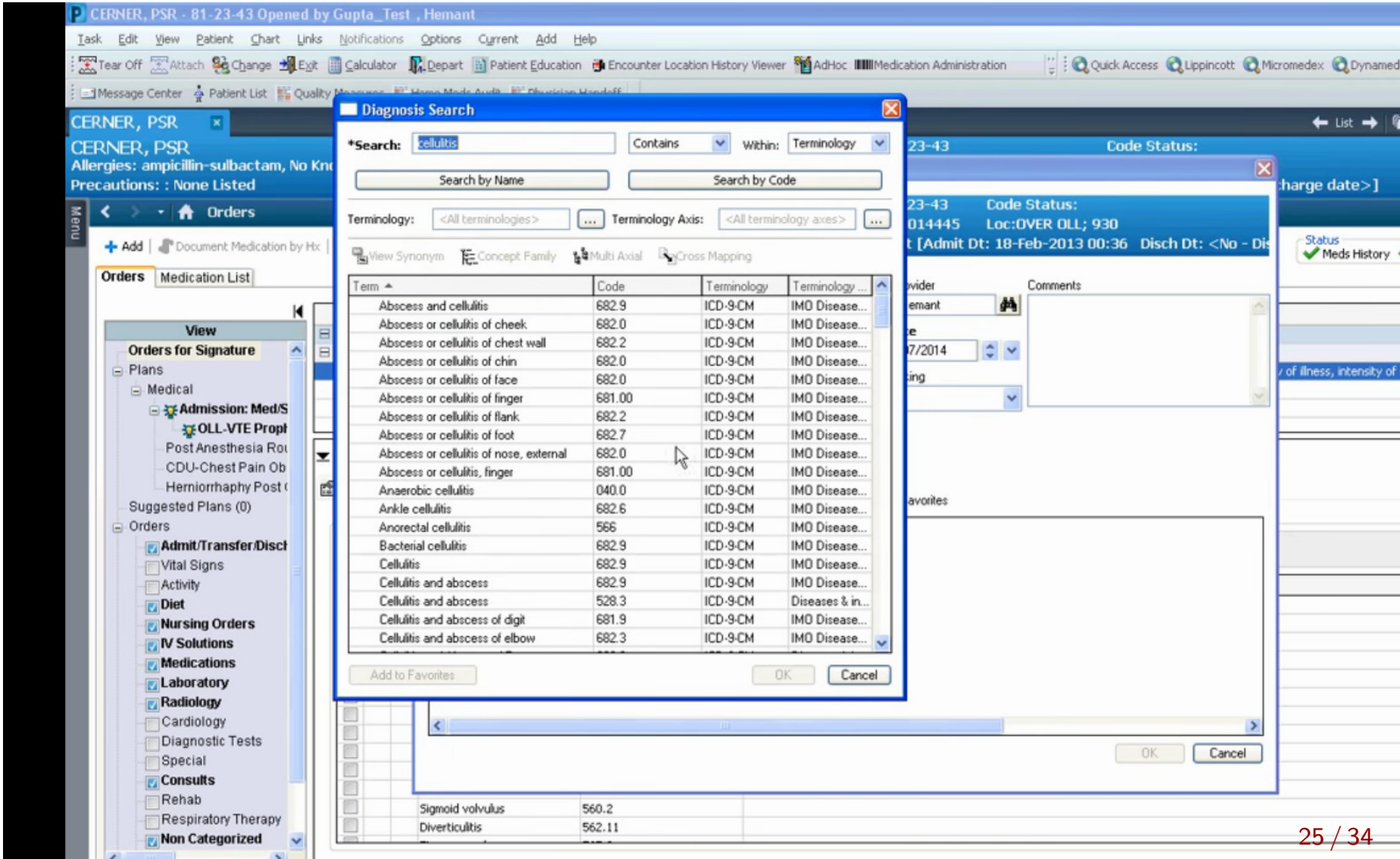
However...

- consultation $\rightarrow$ exam-request
- exam-request: {ECG, allergy test}
- cardiologist-consultation $\rightarrow$ allergy test

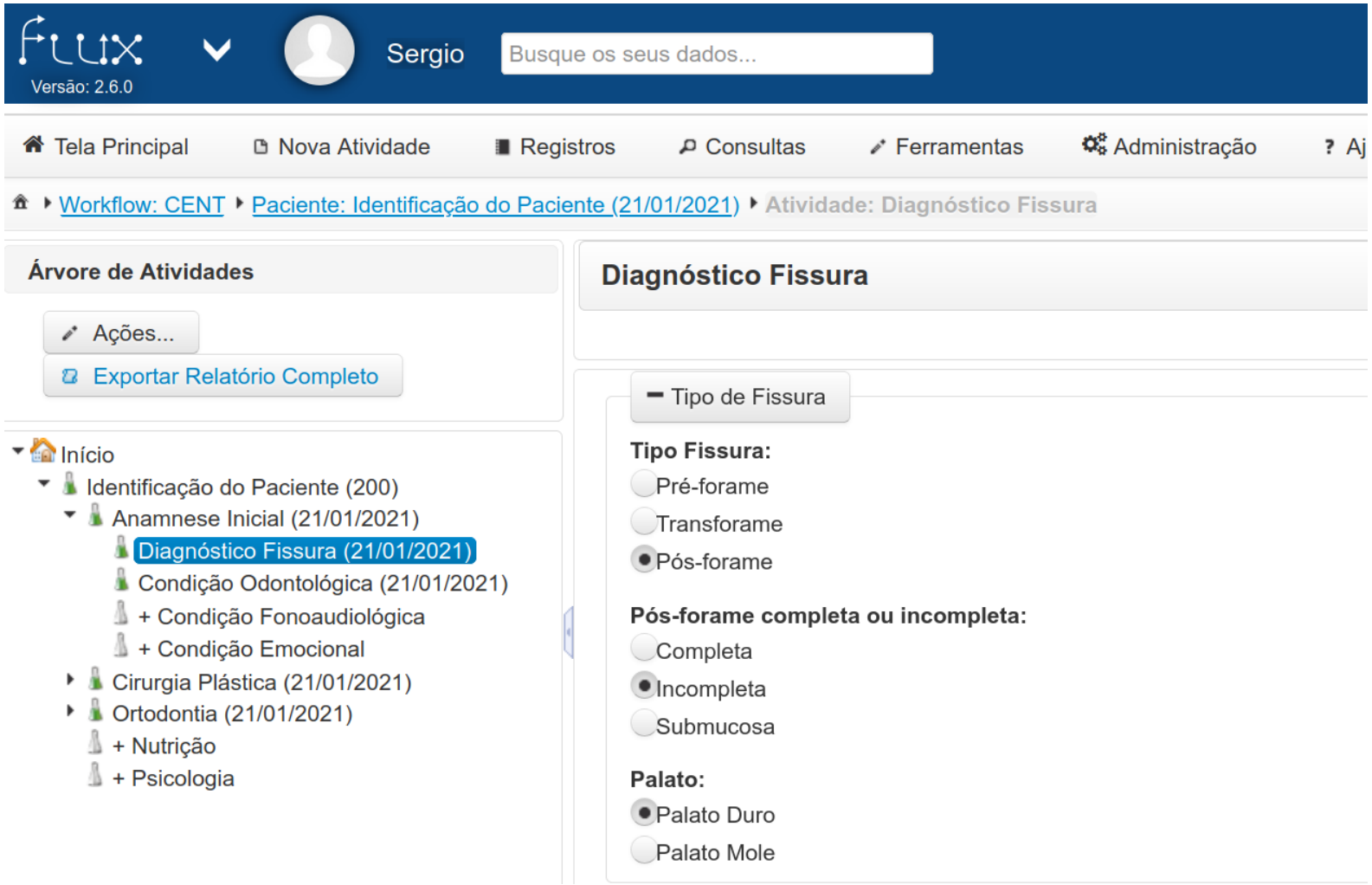
in this example exam-request serves multiple purposes...

Notice the difference...

EPIC Diagnostic Choice



FluxMED Diagnostic Choice



Conceptually Designed EHR: What ?

We can hand tailor the system for each doctor

- Taking into account *context*
  - *Why is it asking me for the tenth time which doctor???!!!*
- Less time filling in data
  - EHRs take away *patient time*

And data is structured:

- Currently doctors data is filled in text fields
- Can't retrieve information such as
  - Which patients are scheduled to return in one month ?
  - How many were diagnosed with X last year ?
  - ...

Conceptually Designed EHR: How ?

Customization is done by adding/removing concepts instances

- For EMS, remove *Schedule*
- For surgery, add *Procedure*
- For lab exams restrict to *Request Exam* and *Exam Result*
- For each specialty, offer specialized *Consultation*

Key: *Hypercustomization*

- Only possible with an appropriate language/tool
- Most people run away from customization, it is hard work!

Conceptually Designed EHR: How ?

Won't there be conflicts ? Not if we design concepts correctly

- Instrumentality: one concept per purpose
- Familiarity: same concept for given purpose
- Integrity: composition preserves concepts

Additionally, concepts should not communicate

- Not always possible to avoid
- But must be restricted
- It is made possible because well designed concepts encapsulate funcionalities.

Conceptually Designed EHR: Why ?

Current EHR systems (or any large system, really)

- Are basically database retrieval tools, glorified CRUD systems
  - Functions are: Add/retrieve patient/consultation, add/retrieve exam result

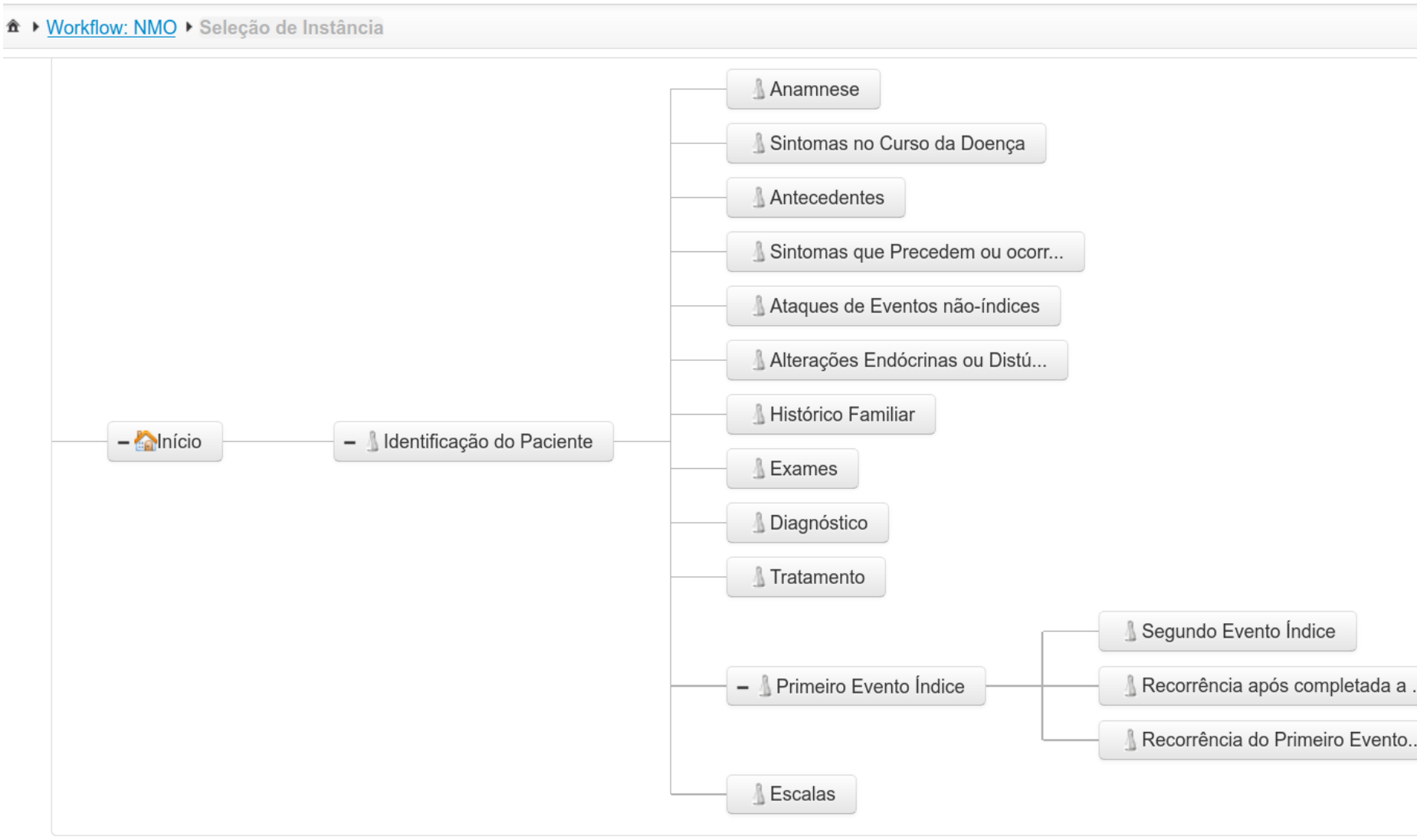
- and not
  - Gather evidence
  - Identify symptoms
  - ...

They center on the database, not on the user...

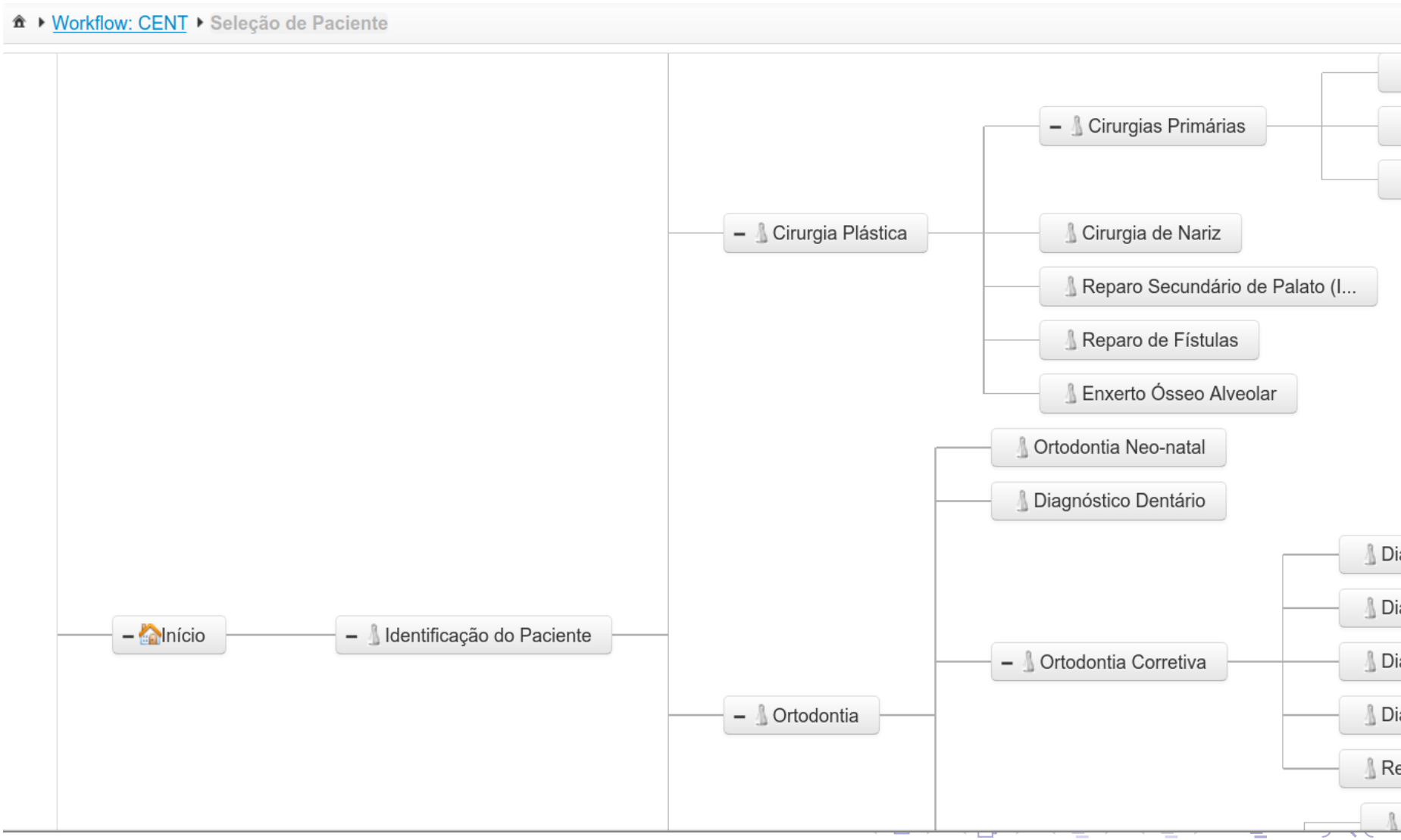
Conceptually Designed EHR: COVID



Conceptually Designed EHR: Neuromyelitis Optica



Conceptually Designed EHR: CENTRARE



Conceptually Designed EHRs

Conceptual design allows us to:

- Isolate functionalities independent of storage
- Combine them to simplify usage
- Minimize communication (source of most problems) in a consistent way

A new paradigm of EHRs that center on users, not Databases.

